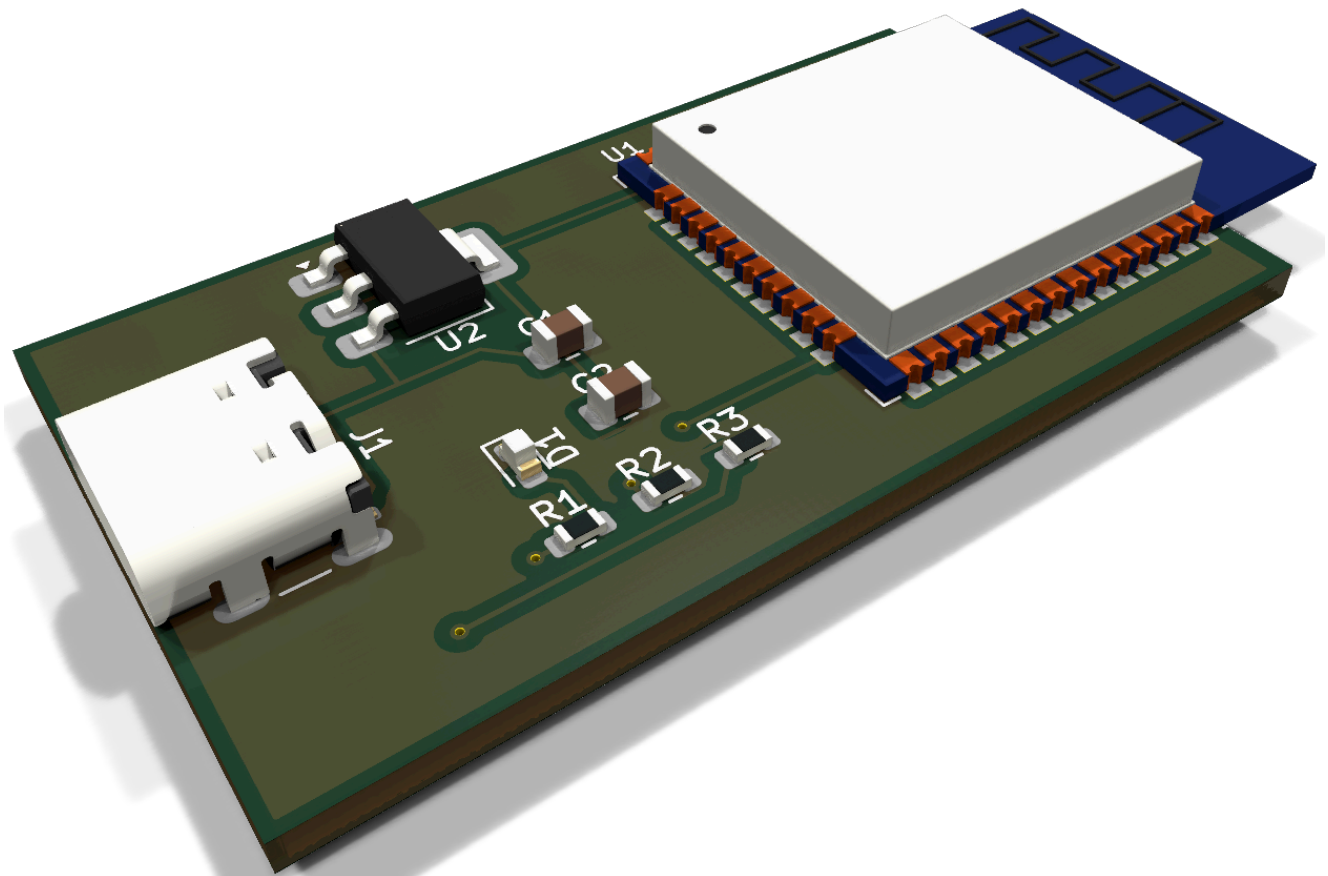


ESP32 + LED + USB-C example board

Hardware design document

Revision A · 2026-09-22 · DEMONSTRATION DESIGN · NOT FOR PRODUCTION



Verification at document generation

ERC	OPEN (2 finding(s))
DRC	PASS (0 open)
Connectivity	0 unconnected
Schematic <-> board parity	0 differences
Reviewed exceptions	4 (justified in section 6)

Generated headless by the kicad-pcb skill. Every verification figure was measured by kicad-cli while this document was built. A clean ERC/DRC report proves rule consistency only: it does not certify electrical safety, RF performance, thermal behaviour or manufacturability. Independent engineering review and the project's release gate are required before ordering hardware.

Document control

Item	Value
Title	ESP32 + LED + USB-C example board
Revision / status	A / DEMONSTRATION DESIGN — NOT FOR PRODUCTION
Generated	2026-09-22 by the kicad-pcb skill (design_doc.py)
Tools	KiCad 10.0.6 (kicad-cli)

Source files

File	SHA-256 (first 16)
esp32-devboard.kicad_sch	59edda6c749d1a5d
esp32-routed.kicad_pcb	672e4be3ee711353

The hashes tie this document to the exact design files it was generated from. Regenerate the document after any change.

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1. Design description and theory of operation
2. 3D schematic
3. Schematic
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8. Open issues and limitations

1. Design description and theory of operation

Purpose and scope

A minimal board used to demonstrate the kicad-pcb skill end to end, headless, on Linux and Windows: schematic generation, schematic-driven board nets, placement, Freerouting, copper pours, DRC with schematic parity, 3D renders and this document. It is **not** a usable ESP32 development board: programming access, reset/boot circuitry and input protection are deliberately missing (see Open issues).

Requirements

ID	Requirement	Source	Status
R1	Powered from a USB-C source, 5 V, default USB current	demo scope	implemented (sink only)
R2	3.3 V rail for an ESP32-WROOM-32 module	module datasheet: 3.0–3.6 V	implemented (AMS1117-3.3)
R3	One user LED on a GPIO	demo scope	implemented (IO2)
R4	2-layer board, 45 x 25 mm, antenna at the board edge	demo scope	implemented
R5	Firmware programming and reset	normal ESP32 practice	not implemented

Architecture

USB-C receptacle J1 (VBUS, CC1/CC2, GND) → +5V → AMS1117-3.3 LDO U2 → +3V3 → ESP32-WROOM-32 module U1. IO2 of U1 drives LED D1 through R1. There is no data path: USB D+/D– and SBU pins of J1 are not connected.

Theory of operation

USB-C power input (J1, R2, R3)

R2 and R3 (5.1 kΩ) pull CC1 and CC2 to ground. These are the Type-C sink termination resistors Rd: a Type-C source detects them on whichever CC line the plug orientation connects and only then enables VBUS. The board draws "default USB power" (500 mA from USB 2.0 ports, 900 mA from USB 3.x ports); it does not read the advertised current level. C1 (10 μF) is the local bulk capacitor on +5V.

3.3 V regulation (U2, C1, C2)

U2 is a fixed 3.3 V linear regulator. Input 5 V, output 3.3 V: the headroom of 1.7 V exceeds the AMS1117 dropout (about 1.1–1.3 V at its 800 mA rating) with margin. C2 (10 μF) is the output capacitor next to the module supply pin.

ESP32 module (U1)

The module contains the ESP32, flash, crystal and PCB antenna; it needs only 3.3 V, ground, an enable (EN) signal and strapping-pin states at reset. VDD (pin 2) is on +3V3, the GND pins (1, 15, 38 and the thermal pad 39) on GND.

LED (IO2, R1, D1)

IO2 (module pin 24) drives the anode of D1 through R1 = 1 kΩ; the cathode goes to GND, so the LED is lit when IO2 is high. IO2 is a strapping pin: it must be low or floating for the serial bootloader. The LED path only conducts above the LED forward voltage, so at reset the pin is effectively floating — compatible.

Design calculations

Quantity	Calculation	Result	Basis
LED current	$(3.3\text{ V} - 2.1\text{ V}) / 1\text{ k}\Omega$	$\approx 1.2\text{ mA}$	Vf 2.1 V assumed for a green 0603 LED (part TBD)
LDO dissipation, typical	$(5 - 3.3)\text{ V} \times 0.08\text{ A}$	$\approx 0.14\text{ W}$	80 mA average assumed (Wi-Fi idle/modem-sleep mix)
LDO dissipation, TX peaks	$(5 - 3.3)\text{ V} \times 0.5\text{ A}$	$\approx 0.85\text{ W}$	Espressif asks for a supply able to deliver $\geq 500\text{ mA}$
Junction rise at 0.85 W	$0.85\text{ W} \times 60\text{--}90\text{ }^{\circ}\text{C/W}$	$\approx 50\text{--}77\text{ }^{\circ}\text{C}$	θ_{JA} of SOT-223 on a small 2-layer board: assumed range, TBD

The peaks are short bursts, so the average dissipation is what heats the LDO; sustained high-duty transmission would need either more copper under the tab or a switching regulator.

Component choices

- **ESP32-WROOM-32**: pre-certified module with PCB antenna, so the demo needs no RF design.
- **AMS1117-3.3**: ubiquitous, cheap fixed LDO; adequate for a USB-powered demo.
- **GCT USB4105 (16-pin USB-C)**: official KiCad footprint and 3D model; the matching symbol is `USB_C_Receptacle_USB2.0_16P` (pin-for-pad).

PCB layout notes

- 2 layers, 45 x 25 mm, 1.6 mm. USB-C opening flush with the left edge; the module sits at the right edge with its antenna past the board edge (no copper under the antenna).
- GND pours on both layers, a stitching via next to every SMD GND pad.
- Signal routing by Freerouting after placement; the pours are added after routing.
- The antenna overhang is documented by a scoped custom DRC rule on U1's silkscreen.

Assumptions and unverified data

- LED forward voltage, exact LED part and all passive part numbers: TBD.
- AMS1117 vendor/datasheet revision: TBD. The original AMS1117 datasheet asks for a 22 μF tantalum output capacitor for stability; C2 is a 10 μF ceramic. Stability with a low-ESR ceramic must be confirmed against the chosen vendor's datasheet.
- θ_{JA} of U2 on this board: not simulated, not measured.
- No part of this board has been built or measured.

Open issues

1. **No programming access**: the original ESP32 has no USB peripheral and UART0 (TXD0/RXD0) is not brought out. Add a USB-UART bridge on the USB-C data pins, or at least a UART header.
2. **EN has no RC**: add 10 k Ω pull-up to +3V3 and 1 μF to GND (Espressif reference design), plus a RESET button. ERC keeps this pin flagged on purpose.
3. **No BOOT button on IO0**.
4. **Decoupling**: add 0.1 μF next to the module supply pin in addition to C2.
5. **LDO output capacitor** type/value per the chosen AMS1117 vendor (see assumptions).
6. **No input protection**: no ESD/TVS on VBUS/CC, no fuse or reverse-current path.
7. **LDO thermal margin** under sustained Wi-Fi transmission is not verified.

Bring-up and test plan

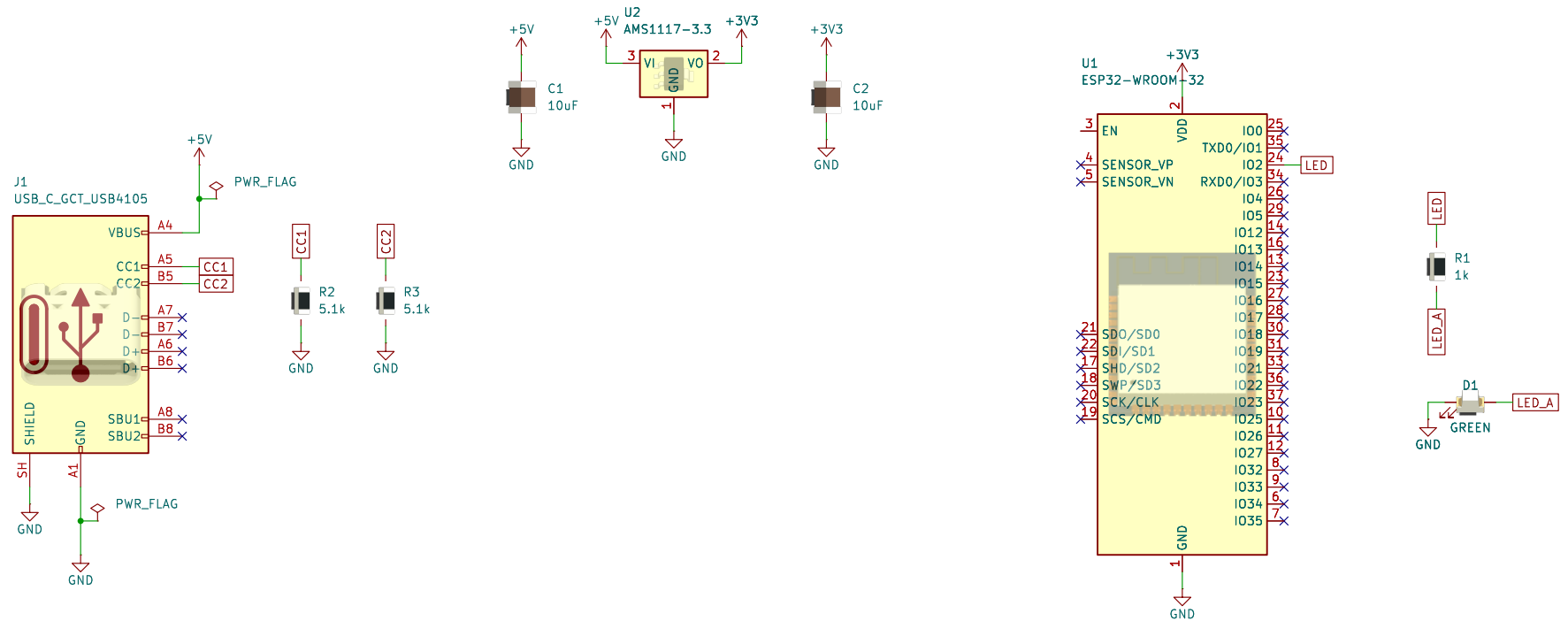
This board is a documentation example; if one is ever built: power from a current-limited 5 V supply (100

mA limit), check +3V3 = 3.3 V $\pm$ 3 %, then raise the limit to 600 mA. Programming requires resolving open issues 1–3 first.

Design facts (extracted from the netlist)

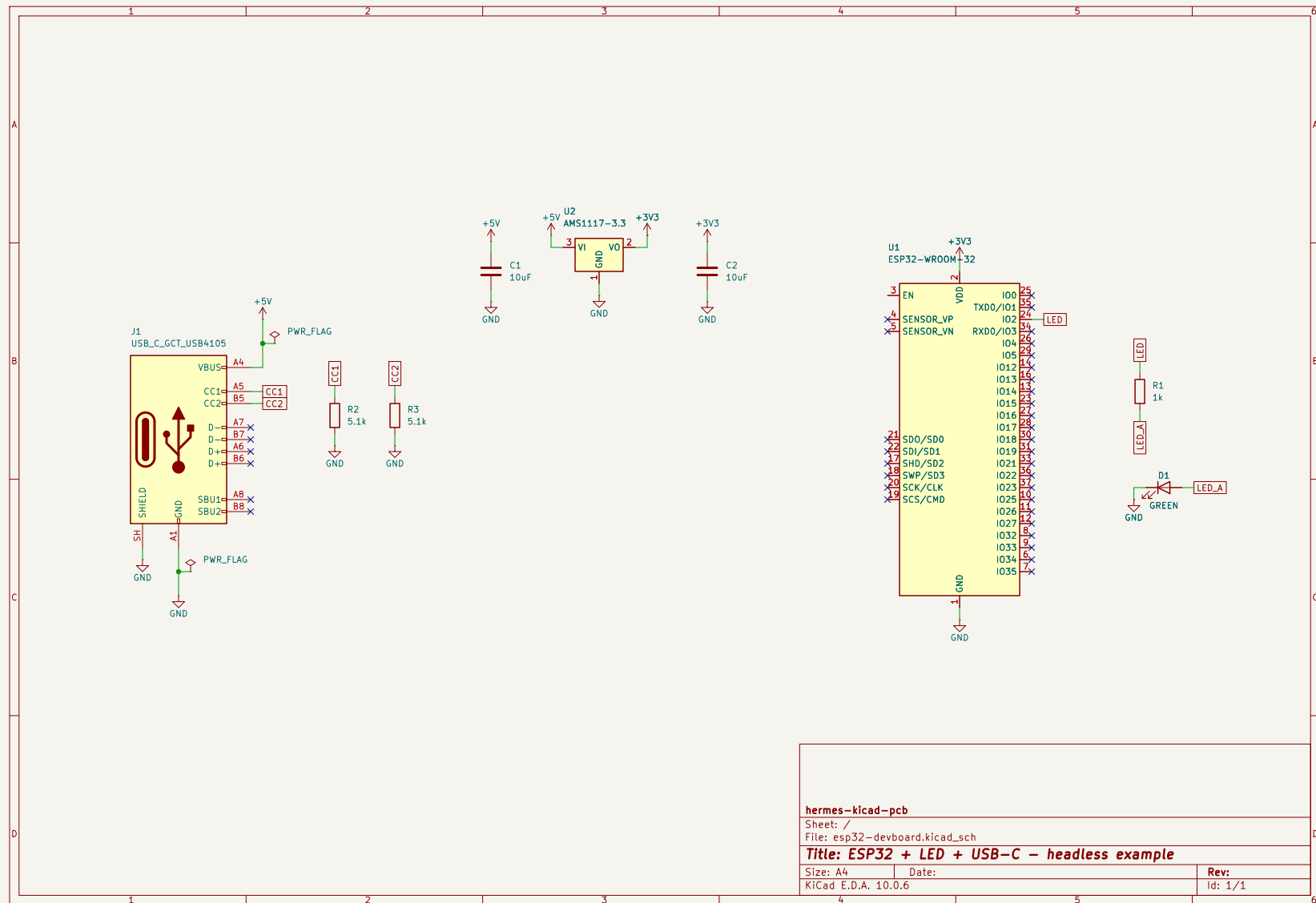
Item	Value
Components	9
Nets	46
Sheets	1
Supply / ground nets	+3V3, +5V, GND

2. 3D schematic · esp32-devboard



Parts drawn with their footprint's 3D model (KiCad renders). Wiring, labels and power symbols are the schematic itself.

3. Schematic

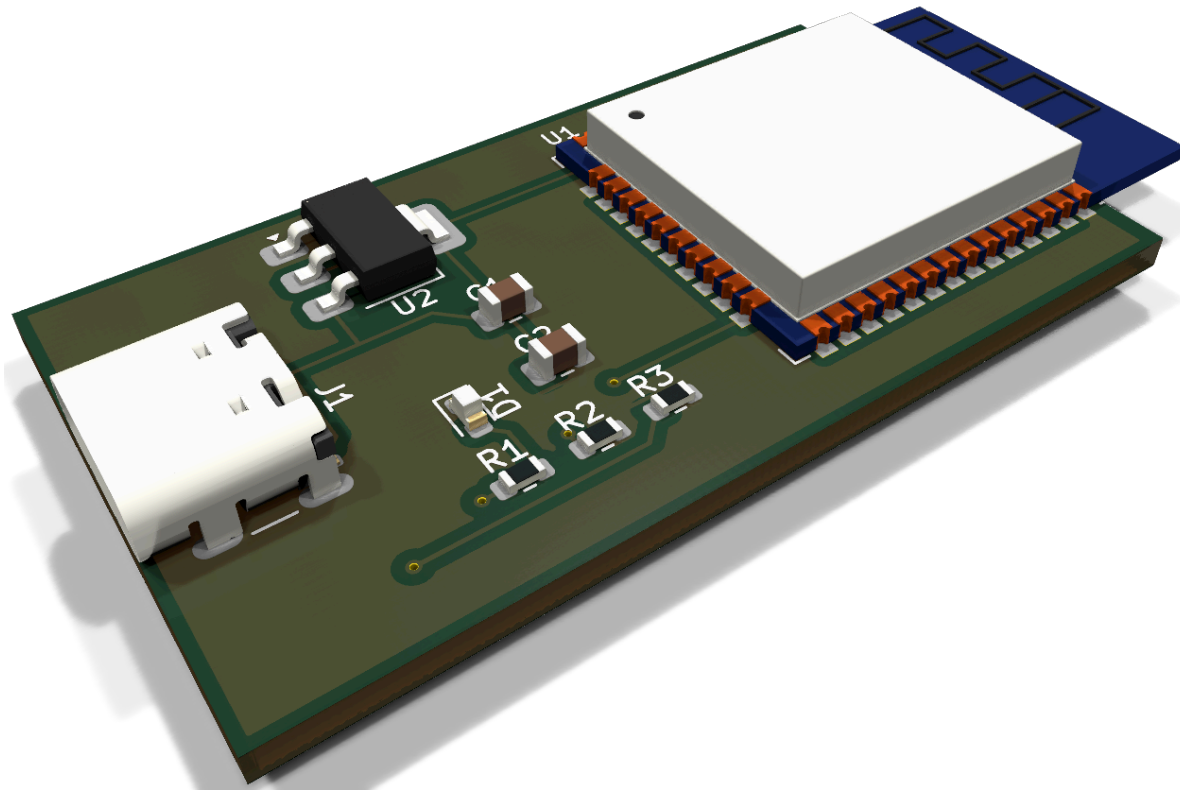


4. Printed circuit board

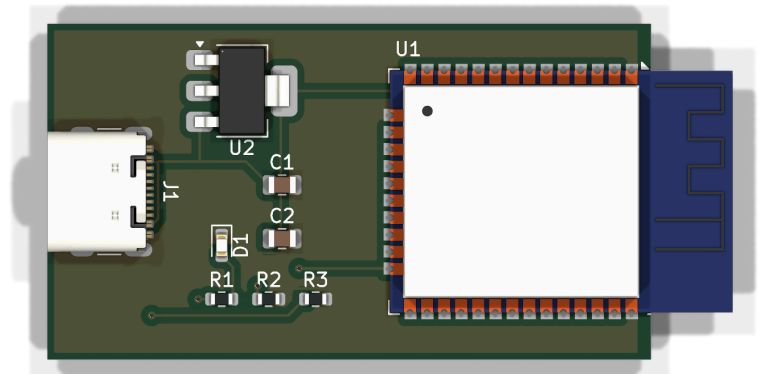
Property	Value
Outline	45.00 x 25.00 mm
Copper layers	2 (F.Cu, B.Cu)
Board thickness	1.6 mm
Footprints	9
Track segments / arcs / vias / zones	89 / 0 / 26 / 3
Narrowest track	0.15 mm

Board views and vector layer plots follow on the next pages (KiCad raytraced renders with the footprints' own 3D models).

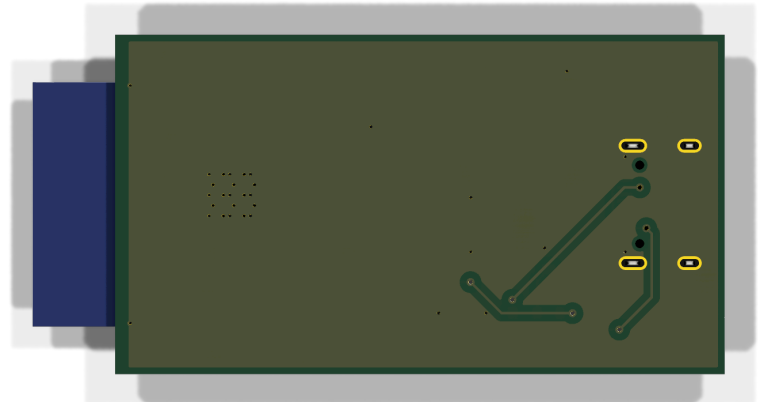
4. Board views (KiCad 3D renders)



3D view

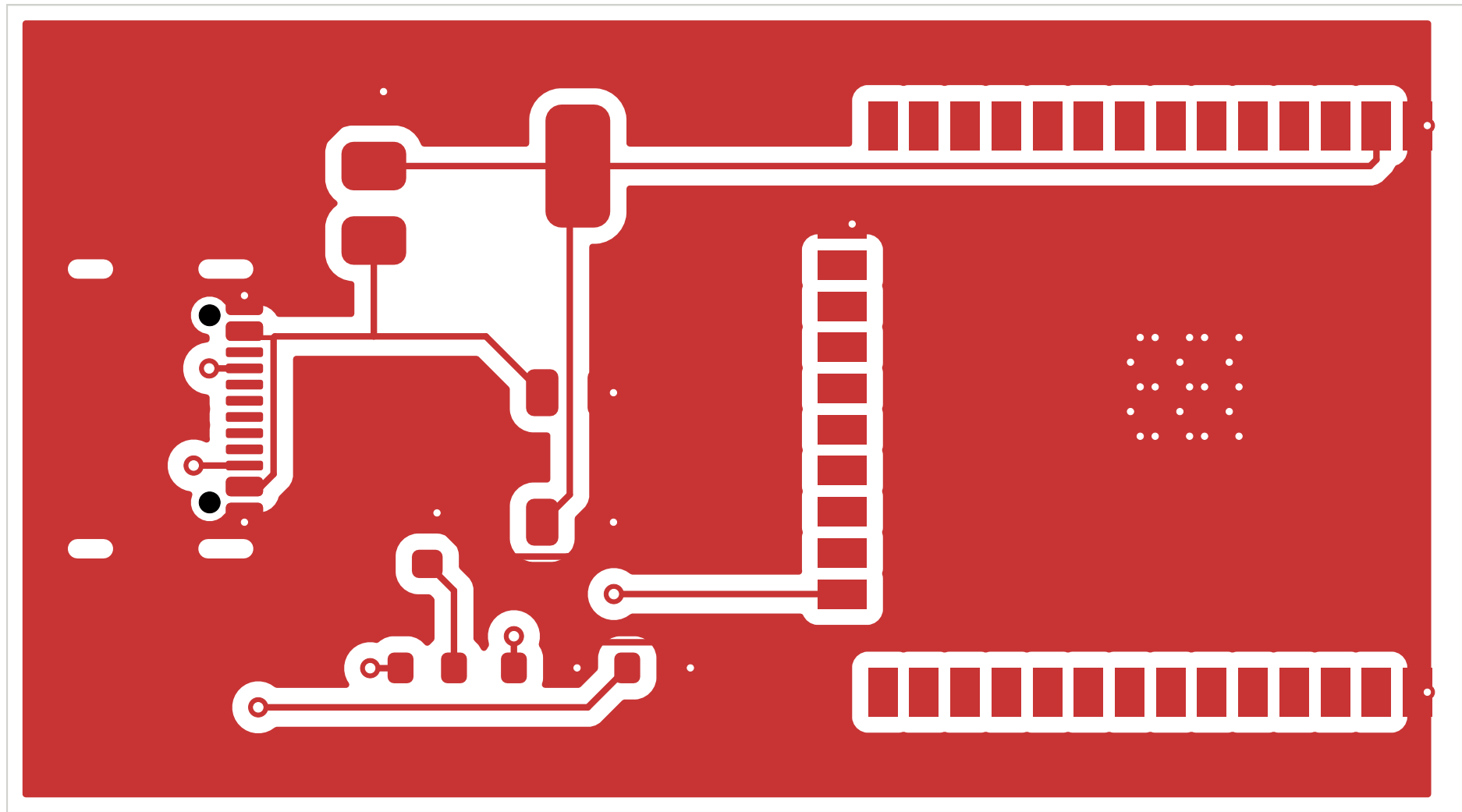


Top side



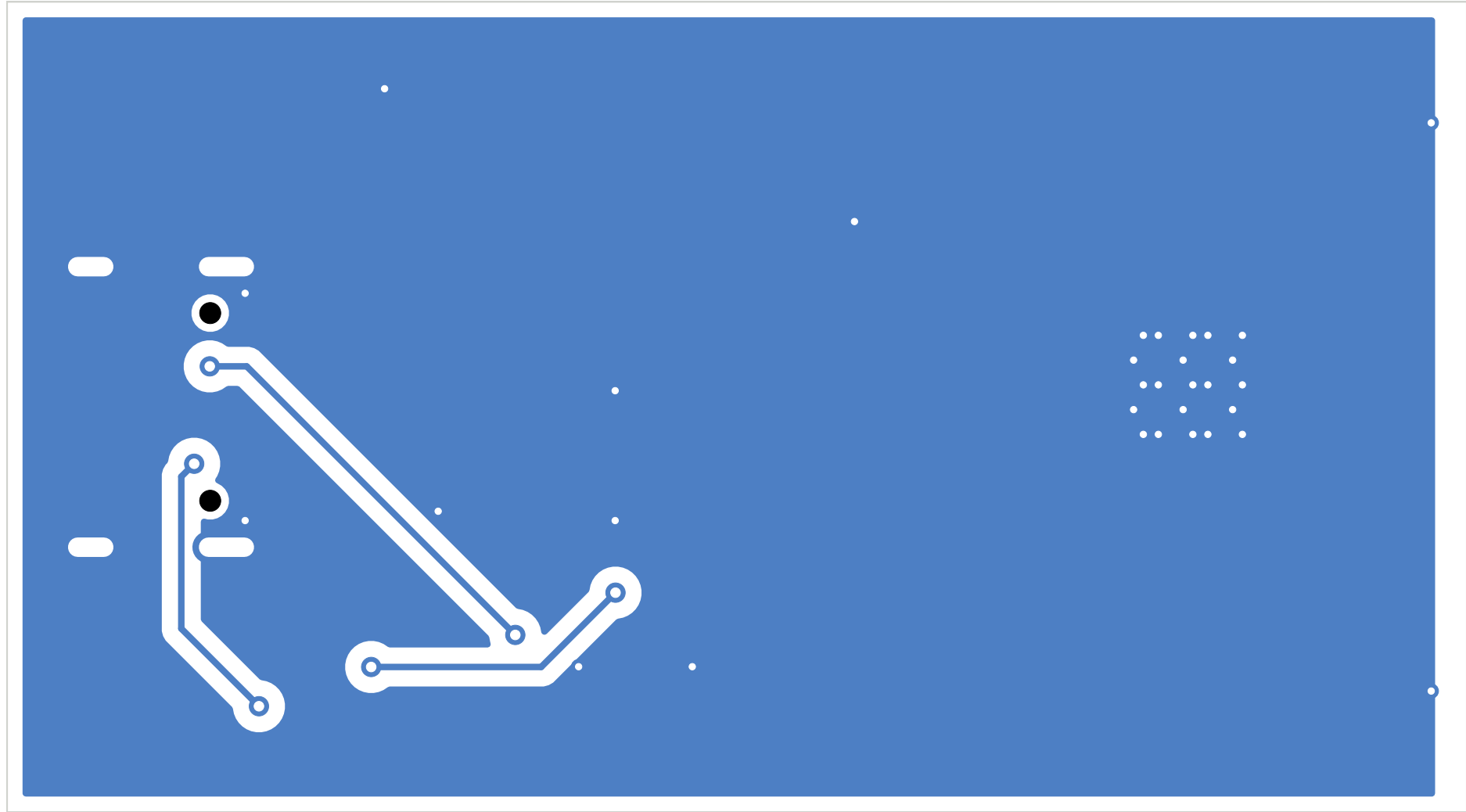
Bottom side

4. Layer plot · F.Cu



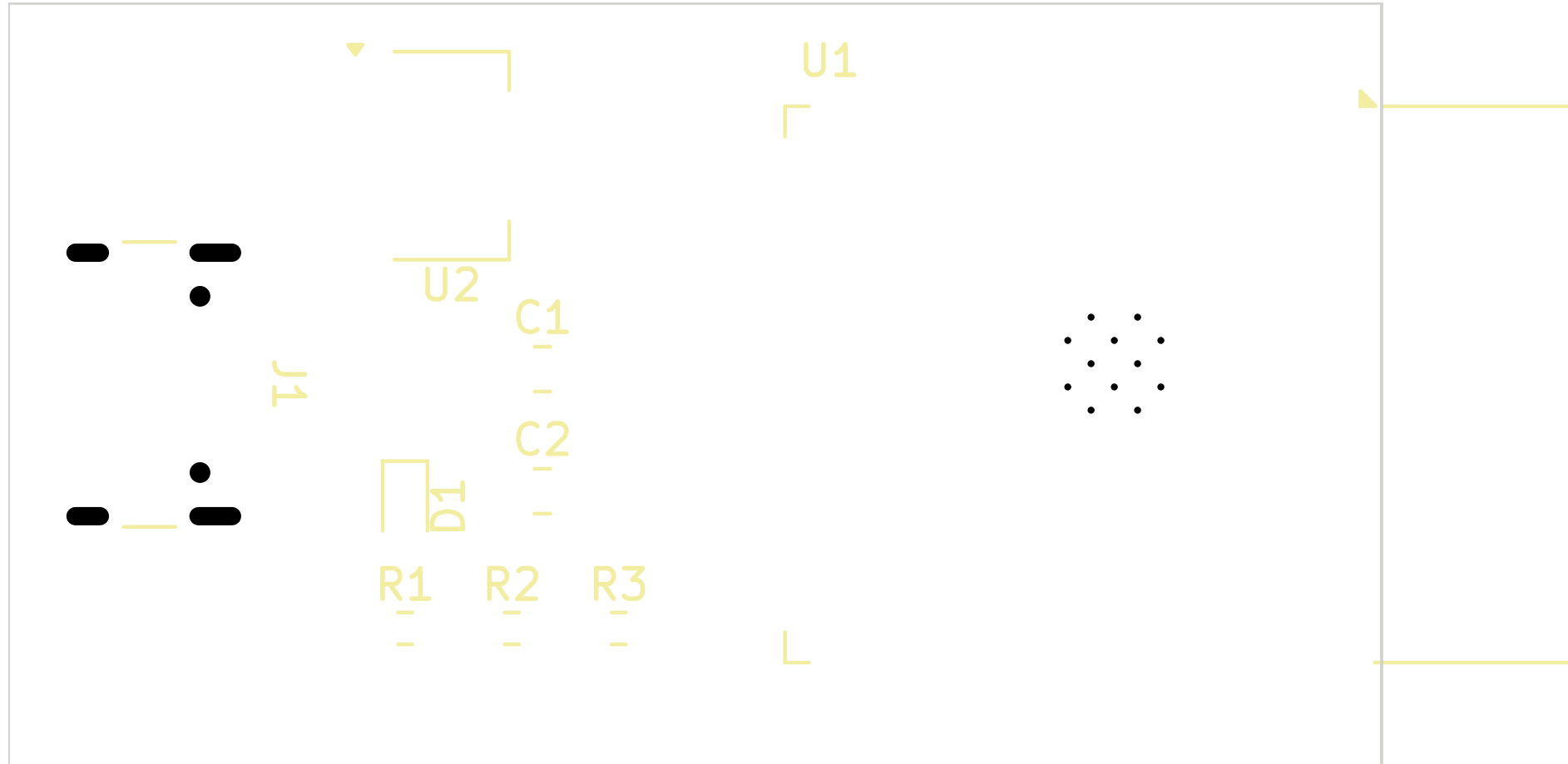
Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · B.Cu



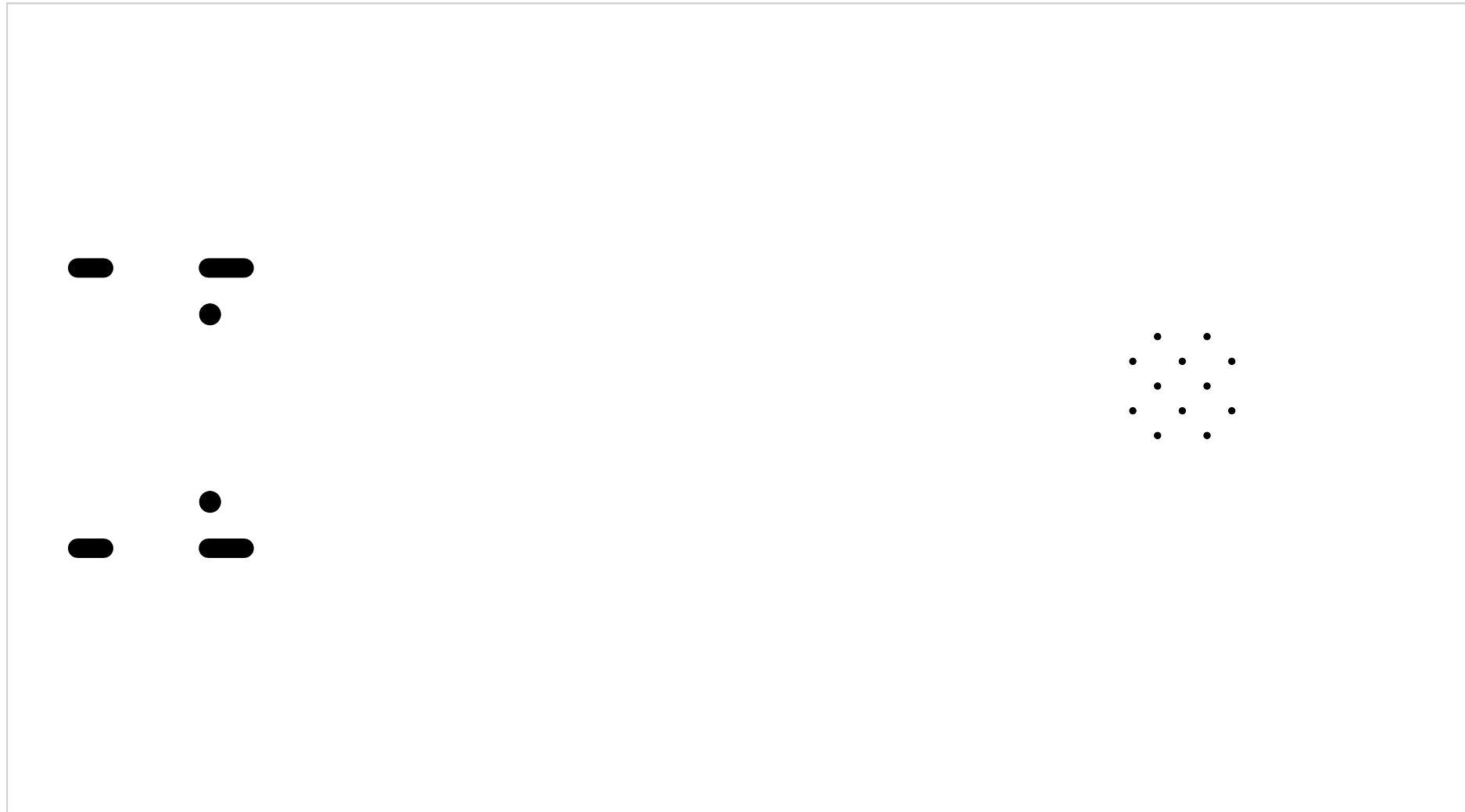
Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · F.Silkscreen



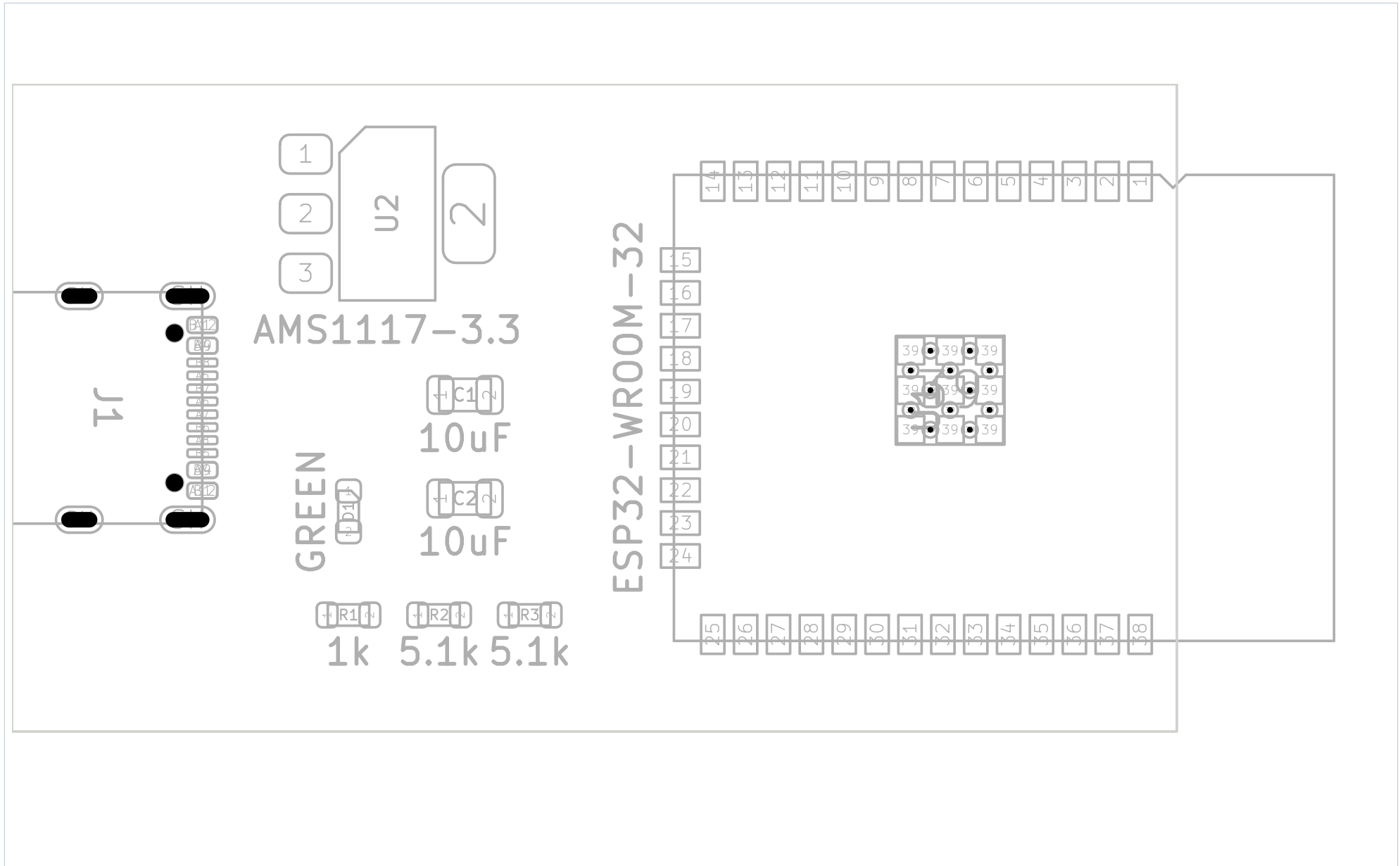
Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · B.Silkscreen



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

4. Layer plot · F.Fab



Vector plot from kicad-cli (board outline on every page); zoomed to the board.

5. Bill of materials

Grouped from the schematic netlist. Supplier columns appear only when the schematic carries the field (LCSC / MPN); no part number or stock class is inferred.

Qty	References	Value	Footprint	Description
2	C1, C2	10uF	C_0805_2012Metric	Unpolarized capacitor
1	D1	GREEN	LED_0603_1608Metric	Light emitting diode
1	J1	USB_C_GCT_USB4105	USB_C_Receptacle_GCT_USB4105-xx-A_16P_TopMnt_Horizontal	USB 2.0-only 16P Type-C Receptacle connector
1	R1	1k	R_0603_1608Metric	Resistor
2	R2, R3	5.1k	R_0603_1608Metric	Resistor
1	U1	ESP32-WROOM-32	ESP32-WROOM-32	RF Module, ESP32-D0WDQ6 SoC, Wi-Fi 802.11b/g/n, Bluetooth, BLE, 32-bit
1	U2	AMS1117-3.3	SOT-223-3_TabPin2	1A Low Dropout regulator, positive, 3.3V fixed output, SOT-223

6. Verification

Check	Result
ERC	OPEN (2 finding(s))
DRC	PASS (0 open)
Connectivity	0 unconnected
Schematic <-> board parity	0 differences
Reviewed exceptions	4 (justified in section 6)

ERC (all severities)

Severity	Type	Count	Example
error	pin_not_connected	1	Pin not connected — Symbol U1 Pino 3 [EN, Input, Line]
error	pin_not_driven	1	Input pin not driven by any Output pins — Symbol U1 Pino 3 [EN, Input, Line]

DRC (all severities)

No open DRC findings.

Unconnected items

None.

Schematic parity

Board matches the schematic.

Reviewed exceptions

Type	Count	Justification (recorded by the reviewer)
hole_clearance	4	J1 = GCT USB4105 vendor footprint: its NPTH alignment pegs sit 0.194 mm from its own GND pads by manufacturer geometry; 0.25 mm is this demo project's default, not a fab limit. Confirm against the chosen fab's NPTH-to-copper capability before ordering.

ERC/DRC measure rule consistency against the project's design rules. They do not replace the engineering review of power, RF, thermal and mechanical behaviour listed in section 8.

7. Manufacturing outputs

No fabrication archive was supplied to this document.

Check every output against the selected fabricator's current capabilities ([references/jlcpcb-rules.md](#) for the JLCPCB class) before ordering.

8. Open issues and limitations

1. **No programming access:** the original ESP32 has no USB peripheral and UART0 (TXD0/RXD0) is not brought out. Add a USB-UART bridge on the USB-C data pins, or at least a UART header.
2. **EN has no RC:** add 10 k Ω pull-up to +3V3 and 1 μ F to GND (Espressif reference design), plus a RESET button. ERC keeps this pin flagged on purpose.
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5. **LDO output capacitor** type/value per the chosen AMS1117 vendor (see assumptions).
6. **No input protection:** no ESD/TVS on VBUS/CC, no fuse or reverse-current path.
7. **LDO thermal margin** under sustained Wi-Fi transmission is not verified.

Detected automatically

- 2 open ERC finding(s) — see section 6.

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